



Covariate effect (param ~ covariate . shape)

	nlmixr2-SCM																nlmixr2-VAE															
CL~BW.power	-1	-1	1	0	-1	-1	0	0	1	2	-2	-4	4	0	1	2	1	0	0	2	2	0	2	2	2	1	-5	-6	2	0	0	-1
CL~BW.lin	0	0	0	0	0	0	0	0	1	-1	3	-1	-3	-1	0	1	3	2	2	0	1	2	3	1	2	4	1	-4	-1	1	2	3
CL~BMI.power	0	-1	1	-2	0	-2	-1	2	-1	2	1	-3	-1	-2	2	-2	1	0	1	1	1	0	0	2	1	-2	-2	0	2	1	3	0
CL~BMI.lin	0	0	0	0	0	0	0	-1	1	1	-2	1	5	0	-1	0	1	3	3	2	2	0	4	0	3	0	4	2	6	6	2	1
CL~CRCL.power	-2	-2	-1	0	1	1	-1	4	-1	-1	0	-1	12	11	11	12	0	1	2	2	1	-2	2	-2	1	0	3	3	-1	1	-4	-1
CL~CRCL.lin	-1	0	0	0	1	2	0	5	0	0	0	0	10	12	9	9	2	2	3	5	1	-1	3	0	2	4	1	1	0	3	0	1
CL~SEX.cat	0	0	-1	0	0	0	0	0	1	2	-1	1	1	0	2	0	2	2	2	4	4	2	3	2	6	7	3	6	6	5	5	2
CL~RACE.cat	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	0	3	3	4	1	3	2	1	6	2	3	4	2	1	4	2
VC~BW.power	-2	0	-3	-16	0	0	-2	-9	-1	-1	1	-5	-1	1	-1	-6	-2	1	20	21	-1	1	12	19	0	2	10	11	4	2	10	13
VC~BW.lin	0	0	-3	0	-1	0	-2	0	0	0	1	0	0	-2	0	3	1	0	16	33	-1	-1	11	23	2	1	11	19	3	2	10	19
VC~BMI.power	-2	-2	1	-2	-2	-1	0	1	-4	0	-1	1	0	0	-1	1	1	0	4	0	-1	0	2	3	-1	0	-2	-1	3	3	1	3
VC~BMI.lin	0	0	0	1	0	0	1	-2	0	0	1	-2	0	0	0	-1	1	0	1	3	0	2	4	1	1	1	3	-2	-1	1	-1	-2
VC~CRCL.power	-1	-2	0	-1	-3	-3	0	2	0	-4	-1	0	0	0	-1	-1	1	-2	2	2	0	-1	1	3	0	-2	-2	0	4	3	-1	-1
VC~CRCL.lin	0	0	1	0	0	0	0	-1	0	0	0	1	0	0	1	0	-1	2	1	2	0	1	3	1	0	0	2	3	-1	3	1	2
VC~SEX.cat	0	0	0	1	0	0	0	0	0	1	1	1	0	0	0	2	2	-1	2	-2	0	-2	2	-2	-1	0	3	-1	1	1	2	-1
VC~RACE.cat	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	-1	0	0	1	1	2	1	2	0	4	2	1	1	2	0	2	1
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Simulation scenario																																

Difference in error rate vs PsN-SCM (linCmt, N = 80). Red = method worse, blue = better, white = matches PsN-SCM.